

Getting UDT1.1 (Fixed version)

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Summary: The below is a way to obtain in-the-limit UDT1.1 behavior. It can seem hacky or ad hoc, but I think it's a good start.

Idea: For further and further nodes in the decision tree, we condition $\mathbb{P}$ on larger and larger policies (instead of single actions). For example, for the first 10 rounds we just condition on single actions as usual, but then, from round 10 to round 20, we condition on all possible conjunctions of actions taken in those rounds (across all possible worlds). And then from 20 to 120, etc. Given some assumptions about

1. not getting infinitely screwed by early actions¹ (catastrophe-free),
2. and not being "so unlucky" that time and time again the important entanglements land between the different unconnected policy regions, despite these boundaries having density 0,

this increasing time horizon will not be a problem.

We still have to solve (by using these increasing policy horizons) the coordination problem. Imagine we face the canonical example: our agent will see 1 or 2, and output A or B. If the two counterfactuals output different letters, they win. Imagine further that both agents in both counterfactuals (or equivalently, the agent prior to counterfactual splitting) completely understand the structure of the problem, so that they know A-B and B-A are exactly as good, and A-A and B-B exactly as bad. How do they coordinate? Well, if as in our framework both agents (or equivalently, the precursor agent) have complete access to the same representation of the decision tree, they can just choose the lexicographically first policy (as UDT1.1 does).

The problem appears when, as happens in the real world, the agents only receive observations, and have to build their representations of decision trees. A same situation can be represented by different decision trees, for example, one where the 1 counterfactual is left, and another where it

¹Although this no longer makes sense if we use time-penalized finite utility for our infinite decision tree, right? Any action can just lock away finitely much utility. Maybe we should say "it doesn't lock away infinitely much utility BEFORE time-penalization", but that feels unnatural, and is not translatable to the more natural case in which, instead of taking an infinite tree and time-penalizing it, we just assume that each infinite branch sums to finite utility (and maybe also something like "these finite values are bounded"). Maybe the conditional result to be proven here should just read "for decision trees in which an arbitrarily high fraction of the total (finite) utility is arbitrarily far away from the earliest nodes, we get arbitrarily few coordination failures".

is right. If we could ensure the agents share the same representation of this isomorphic decision trees (that is, they pick the same representative from the reference class), they could just choose the lexicographically first policy. But this (choosing the same representative tree) is easy to obtain: say that the precursor agent (or equivalently, the UDT prior) has settled on a method to choose a representative. For example, they use a lexicographical order on *observations* (like the bits on a picture), and use that to decide between decision trees (something like "the lexicographically first observations go as to the left of the tree as possible").

Realistic scenarios: This does assume a certain cartesian immutability of the input channels (for example, two nerves from my retina to my brain don't randomly swap places), but we're probably okay granting that for now.

On a different note, it of course might be the case that the two counterfactual agents have different opinions about the situation (not only their representations). For example, maybe one of the two has most of its probability mass on a completely different decision tree. These are unavoidable problems of indexical uncertainty (the agent might think it's very unlikely to be in this situation, and correctly ignore, and yet it sometimes happens), and the agents have to reason about these complications themselves ("I would cooperate here, but in this counterfactual I don't cooperate because it's just too unlikely to be in this decision tree and the payoff is not high enough, so I'll just not cooperate"). The idea above just fixes "all there is to fix".

Might this be fragile on implementation? Agents will also be uncertain about the round number. And depending on the round number, the agent can or cannot coordinate with this or that agent (because we are optimizing over smaller or larger policy chunks). But I think this uncertainty is in no way more worrisome than the above general empirical uncertainty (it's just one more piece of it). And additionally, this worry in particular should decrease for latter rounds, since there boundaries between policy regions become rarer, and so it's less likely to mess up "which policy size should I be optimizing now".

Can we nonetheless do better? $\mathbb{P}$ is an infinite distribution on all logical sentences (despite its information seed being the finite $\mathbb{Q}_k$). Instead of increasing the finite policy horizon as above, what if we just take the infinite policy horizon from the start? We can indeed "condition" $\mathbb{P}$ on an infinite conjunction (of action points), by taking the limit of the conditional probability (about, say, total U obtained) for larger finite initial segments of it. A priori this might not converge... but given our assumption that each branch has a finite amount of utility, PLUS these amounts are bounded, I think it does always converge! (Since arbitrarily far-off nodes must yield an arbitrarily small change in U .) Also, taking an arbitrary limit is not computable, but given the "information that has gone into the prior" is finite, probably some decidable proof system will prove the value of the limit (by just taking into account all of the finite information).

We might worry that this is just "letting the earliest prior decide all actions, without updating in any way, and so this doesn't learn". Again, I think we can edit the prior to hard-code something like "I don't yet know what the best action here will be, but (maybe because I believe strongly that there are no dangerous correlations) it will be at least as updateful as looking at $\mathbb{Q}_n$, and implementing this same procedure of infinite policy selection for the infinite tail of that part of the tree".